

SCRATCH GAME MAKERS

Build Your Catching Game

Join Pixel to code a game with
scores, falling objects, and
keyboard controls!

LEA LABS · SCRATCH FOR KIDS



How to Play

STEP 1

Move the Basket

Use the left and right arrow keys on your keyboard to slide the basket across the stage.

STEP 2

Catch Objects

Watch out for falling stars and apples! Position the basket right underneath them to make a catch.

STEP 3

Earn Points

Every time you catch something, your score goes up. Try to get the highest score possible!

Meet the Game Pieces

THE CATCHER

Usually a basket, bowl, or even a hungry character! This sprite stays at the bottom of the screen.

FALLING OBJECTS

Apples, stars, or anything you want to catch. These sprites fall from the top of the screen.

THE SCORE

A special number that keeps track of how many things you've caught. We call this a "Variable".



Mini Quiz 1: What Starts the Game?

Which block should we use to start our catching game when the player is ready?

A

When  clicked

B

When [space] key
pressed

C

When this sprite
clicked

TEACHER / FACILITATOR GUIDE

Answer: A! The Green Flag is the universal "Start" button for Scratch games. It tells all the sprites to wake up and begin their starting scripts at the exact same time.

Controls: Catch With Arrow Keys

MOVE RIGHT

Use the right-arrow block to move the basket along the X axis.

change x by 10

MOVE LEFT

Use the left-arrow block. To move left, use a negative change value.

change x by -10

WHAT IS X?

X is the horizontal position. Changing X moves a sprite left or right.



Mini Quiz 2: Move the Basket

How do we make the basket move to the RIGHT when we press the right arrow key?



TEACHER NOTE: **ANSWER IS A** (X MOVES LEFT/RIGHT, Y MOVES UP/DOWN)

A

when [right arrow] key pressed

change x by (10)

B

when [right arrow] key pressed

change y by (10)

C

when [up arrow] key pressed

change x by (10)

Make Objects Fall

1. START AT THE TOP

Send the object to the top and pick a random x position.

go to x: (pick random) y: 180

2. FALL DOWN

Use a loop to move it down repeatedly.

change y by -5

3. RESET AT THE BOTTOM

If it hits the bottom edge, send it back to the top.

if <y position < -170> then



Catch It! Use an If Block

THE QUESTION

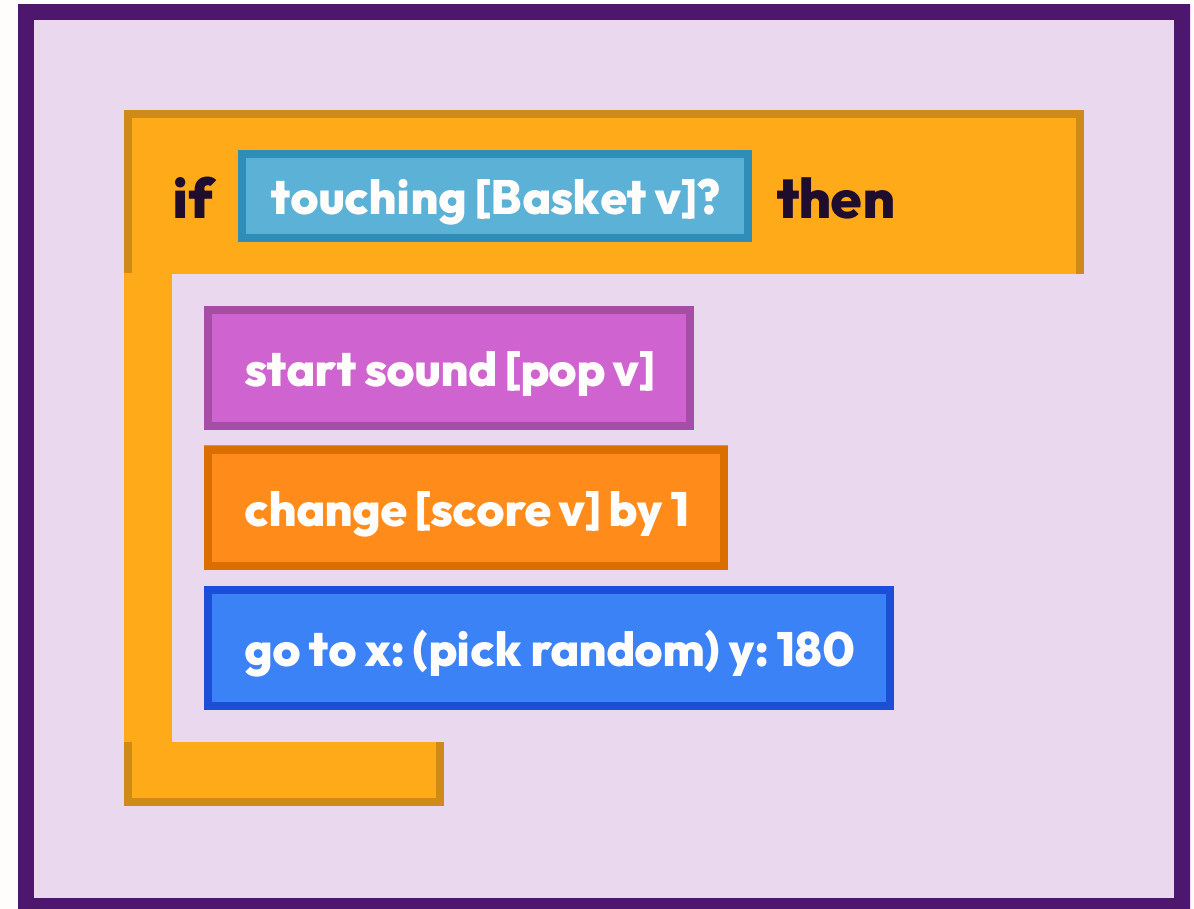
The computer needs to ask: "Is the falling object touching the basket?" We use a light blue **Sensing** block for this.

THE DECISION

Put the question inside an orange **If... Then** block. If YES, the code inside runs.

THE ACTION

When a catch happens, play a sound, add a point, and send the object back to the top to fall again.



Score Points and Track Lives

WHAT IS A VARIABLE?

A variable is like a digital box that holds a number. We use them to remember the player's score and how many lives they have left.

SCORING POINTS

When the basket catches an object, we add 1 to the score variable to reward the player!

change [score v] by 1

LOSING LIVES

If the object misses the basket and hits the bottom edge, we take away 1 life using a minus sign.

change [lives v] by -1



Mini Quiz 3: After a Catch

Inside the "If touching Basket" block, what is the best order for our actions?



TEACHER NOTE: **ANSWER IS A** (WE WANT TO PLAY THE SOUND AND SCORE BEFORE RESETTNG THE POSITION)

A

start sound [pop v]

change [score v] by 1

go to x: (pick random) y: 180

B

go to x: (pick random) y: 180

start sound [pop v]

change [score v] by 1

C

change [score v] by 1

go to x: (pick random) y: 180

start sound [pop v]

Debugging Checklist

1. CHECK THE START

Does the game begin and the score reset to 0 when you click the Green Flag?

2. CHECK THE CONTROLS

Does the basket move left and right smoothly when you press the arrow keys?

3. CHECK THE CATCH

Does the score go up and a sound play when the falling object touches the basket?

4. CHECK THE RESET

Does the object go back to the top of the screen after it is caught or hits the bottom?



Lab Launch: Your Catching Game

DELIVERABLES

- ▶ A catcher sprite that moves left and right using the arrow keys.
- ▶ A falling sprite that drops from a random position at the top of the stage.
- ▶ A "Score" variable that tracks how many objects you catch.

SUCCESS CRITERIA

- ▶ The game starts cleanly when the Green Flag is clicked (score resets to 0).
- ▶ The falling object resets to the top if it hits the bottom edge.
- ▶ A sound plays and the score increases on a successful catch.

EXTENSION CHALLENGE

Add a "Danger" object (like a bug or a rock) that falls too! If your catcher touches it, decrease the score or take away a life.